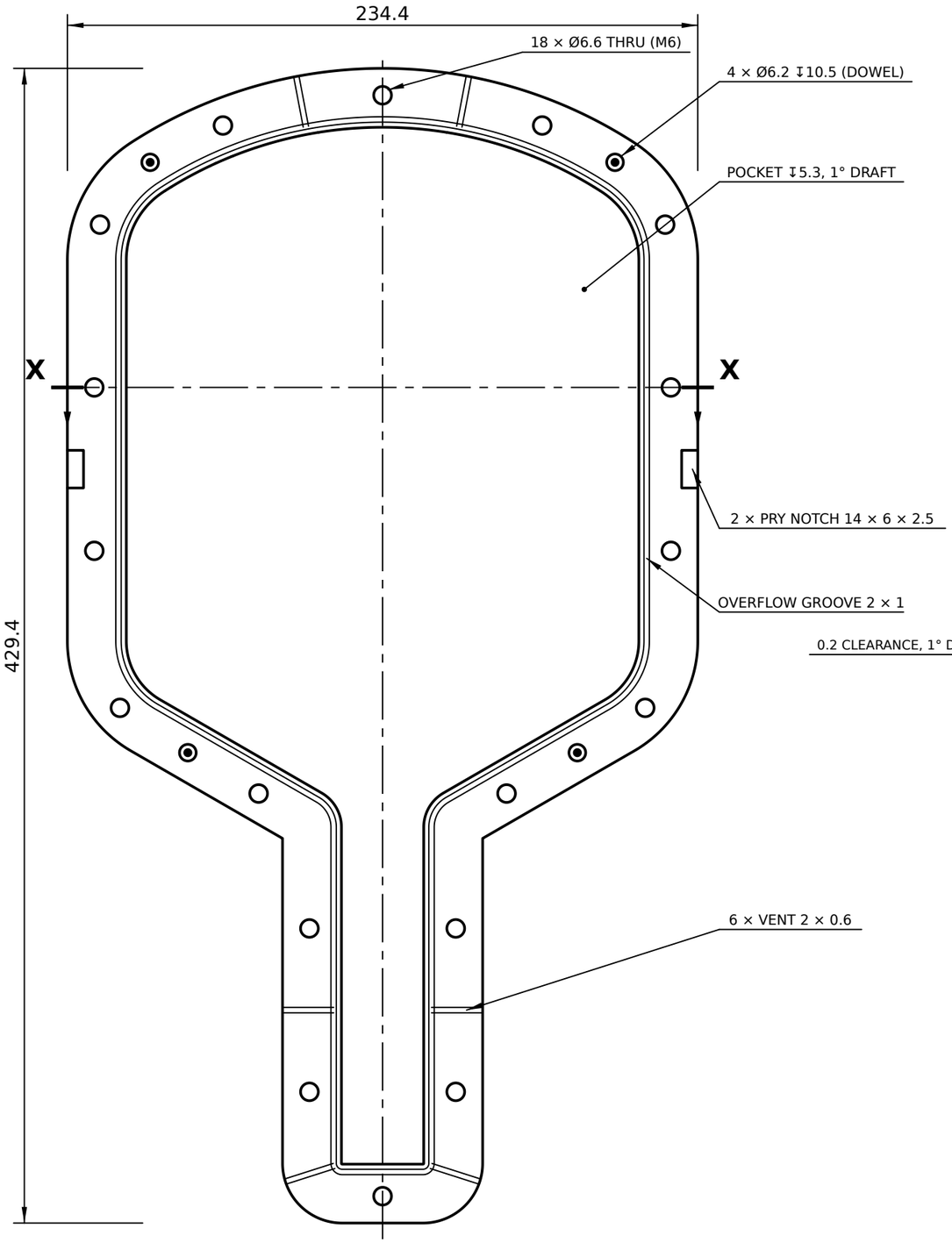
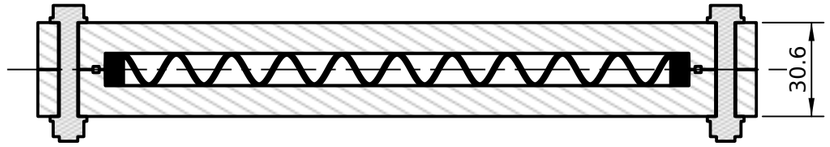


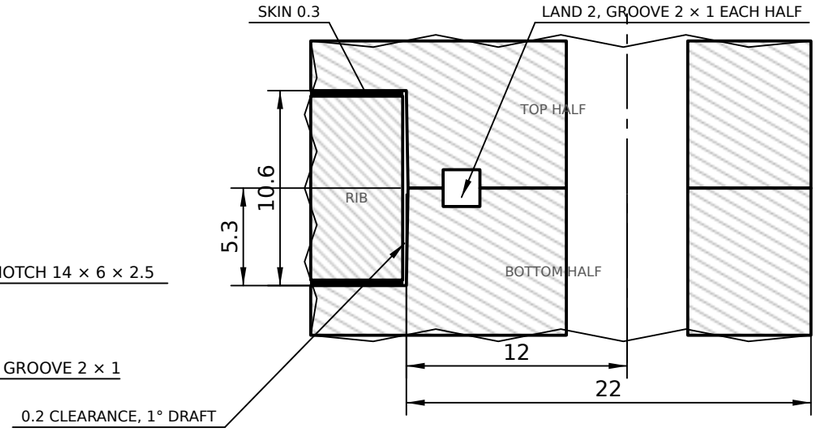
MOULD HALF – PARTING FACE

SCALE 1:2 · BOTH HALVES ARE THIS PART



SECTION X–X

SCALE 1:2 · CLOSED MOULD, SANDWICH 0.3 / 10 / 0.3, M6 BOLT



DETAIL Y

SCALE 3:1 · EDGE OF THE CLOSED MOULD AT X–X

MOULD HARDWARE

ITEM	PART	QTY
1	Mould half, hybrid_mould_half.stl (print 2)	2
2	Hex bolt M6 × 45, 2 washers, nut	18
3	Dowel pin Ø6 × 20, steel (ISO 2338)	4
4	Release wax + PVA, or release film	a/r
5	Flat platens ≥ 18 mm + clamps, or press	2

NOTES

1. MOULD HALF (1): PRINT TWICE – BOTH HALVES ARE THE SAME PART (output/mould/hybrid_mould_half.stl / .step); TURN ONE OVER ONTO THE OTHER. POCKET 5.3 DEEP IN EACH HALF = CAVITY 10.6 = SKIN 0.3 + RIB SUPPORT 10 + SKIN 0.3 WHEN THE FLANGES MEET.
2. PRINT POCKET UP, BACK ON THE BED: ≥ 5 WALLS, ≥ 40 % INFILL, 6 TOP LAYERS + IRONING ON THE POCKET FLOOR. NEEDS A ≥ 440 mm BED; OR MACHINE THE STEP FROM ALUMINIUM / HDPE. ABOUT 1.0 kg OF PLA AT 100 % INFILL.
3. PLA SOFTENS NEAR 55 °C: ROOM-TEMPERATURE-CURE EPOXY ONLY. FOR A WARM POST-CURE PRINT IN PETG, ASA OR PA-CF.
4. PREPARE: SAND THE POCKET FLOORS (P400 → P800), SEAL (EPOXY COAT OR PRIMER), WAX + PVA RELEASE, OR LINE WITH RELEASE FILM. PRESS THE 4 DOWEL PINS (3) INTO ONE HALF.
5. LAY UP IN THE BOTTOM HALF: SKIN (CUT TO THE PROFILE, 0.5 UNDERSIZE), WET OUT → RIB SUPPORT (FACES ABRADED, M3 INSERT FITTED, SCREW IN IT) → TOP SKIN, WET OUT → TOP HALF OVER THE DOWELS. THE POCKET LOCATES THE RIB SUPPORT (+0.2).
6. CLOSE: BOLTS (2) FINGER-TIGHT, THEN PRESS THE MOULD BETWEEN 2 FLAT, STIFF PLATENS (≥ 18 mm PLY OR ALUMINIUM, CLAMPS EVERY ~100 mm), IN A PRESS OR IN A VACUUM BAG UNTIL THE FLANGES MEET; THEN TIGHTEN THE BOLTS. BOLTS ALONE BOW A PRINTED MOULD AND CANNOT PRESS THE MIDDLE OF THE BLADE.
7. EXCESS RESIN FILLS THE OVERFLOW GROOVE (2 × 1 IN EACH HALF) AND LEAVES BY THE 6 VENTS. CURE, SPLIT AT THE PRY NOTCHES, TRIM THE FLASH AT THE EDGE.
8. MEASURE THE RIB SUPPORT (10 ±0.1) BEFORE MOULDING; SAND THE LANDS IF THICK. SHIM THE FLANGE (TAPE / FOIL) TO OPEN THE CAVITY, E.G. FOR PRE-CURED SKINS AND GLUE LINES.

COMPRESSION MOULD – HYBRID

TWO-HALF COMPRESSION MOULD · CAVITY 10.6 mm · 234.4 × 429.4 × 15.3 mm PER HALF

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-24		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-HY-001	D	4 / 4	A3	1:2, 3:1	mm

PROJECTION	MATERIAL	STATUS
	PLA / PETG (NOTE 2)	ISSUED FOR REVIEW